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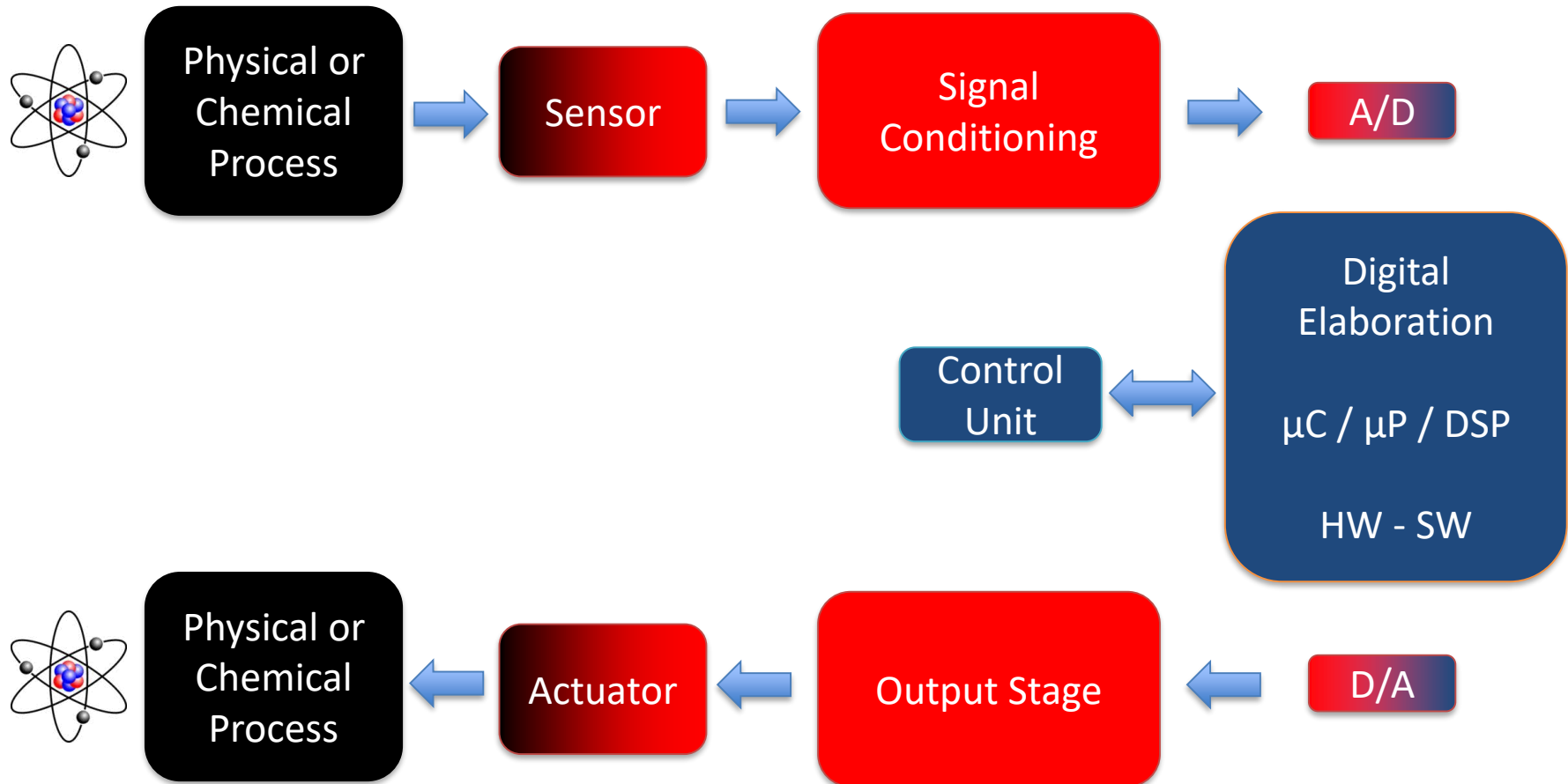
A/D and D/A Conversion

Laurea Triennale in Ingegneria Informatica

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At the edge

We got the signal how we wanted... but what for?



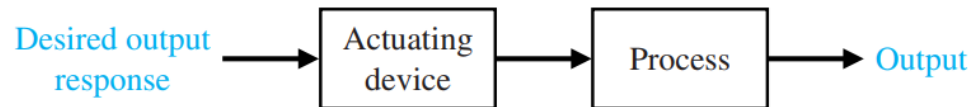
The Core

Control!

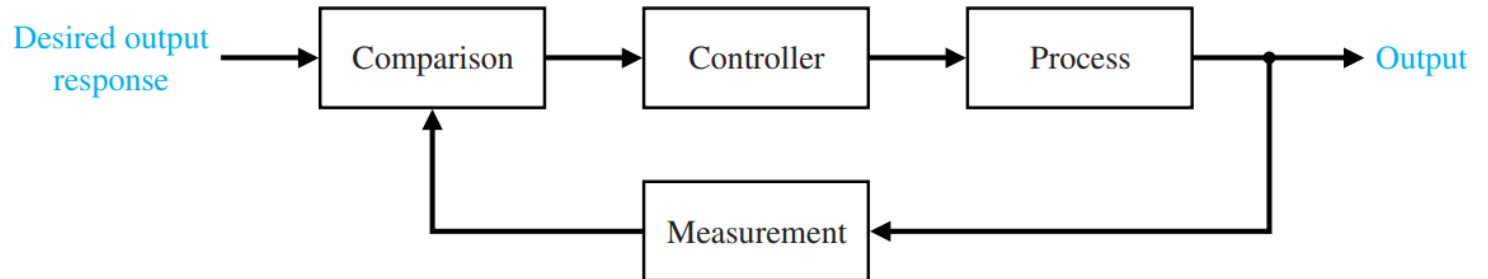
No Control



Feed-Forward
Control (open loop)



Closed-Loop
Control



Why Digital?

Digital has many benefits!

Digital control is a branch of control theory that uses digital computers to act as **system controllers**. Depending on the requirements, a digital control system can take the form of a microcontroller to an ASIC to a standard desktop computer.

Since a digital computer has finite precision (i.e., introduces quantization errors), extra care is needed to ensure the error in coefficients, A/D conversion, D/A conversion, etc. are not producing undesired or unplanned effects.

Since the creation of the first digital computer in the early 1940s the price of digital computers has dropped considerably, which has made them key pieces to control systems for several reasons:

- **Inexpensive**: under \$5 for many microcontrollers
- **Flexible**: easy to configure and reconfigure through software
- **Scalable**: programs can scale to the limits of the memory or storage space without extra cost
- **Adaptable**: parameters of the program can change with time
- **Static operation**: digital computers are much less prone to environmental conditions than analog

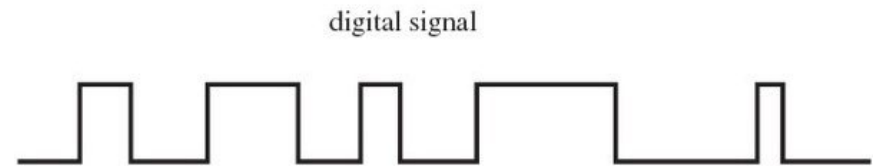
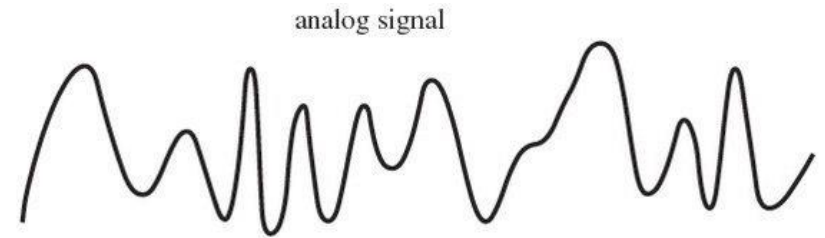
Analog-to-Digital Conversion

ADC

Terminology

Analog: continuously valued signal, such as temperature or speed, with infinite possible values in between

Digital: discretely valued signal, such as integers, encoded in binary



Analog Signals

The world out there is analog

Analog signals – directly measurable quantities
in terms of some other quantity

Examples:

- Thermometer – mercury height rises as temperature rises
- Car Speedometer – Needle moves farther right as you accelerate

Digital Signals

The world of microprocessors is digital

Digital Signals – have only two states.

For digital computers, we refer to binary states, 0 and 1.

1 can be on, 0 can be off.

Examples:

- Light switch can be either on or off
- Door to a room is either open or closed (well, kind of)

How do we go from analog to digital and back?

Well, we are talking about numbers after all so it seems pretty obvious to take such (decimal) numbers and convert them to binary! But this is not enough!

Analog signals are **continuous** sequences of **real numbers**!

- **Continuous**: in a finite chunk of a signal there are infinite points
- **Real numbers**: what about the precision?

A/D and D/A

The devil is in the details

Analog signals are **continuous** sequences of **real numbers**!

- **Continuous**: in a finite chunk of a signal there are infinite points
- **Real numbers**: what about the precision?

Well, it takes no big fantasy to provide an answer to the above points!

Continuity must be broken: **sampling**!

Precision must be limited: **quantization**!

Then we will deal with finite datasets of real numbers with finite precision representation that can be then **encoded** in binary.

Sampling a signal

Taking snapshots

It is a process of taking a sufficient number of discrete values at point on a waveform that will define the shape of waveform.

The more samples you take, the more accurately you will define the waveform.

It converts analog signal into series of impulses, each representing amplitude of the signal at given point.

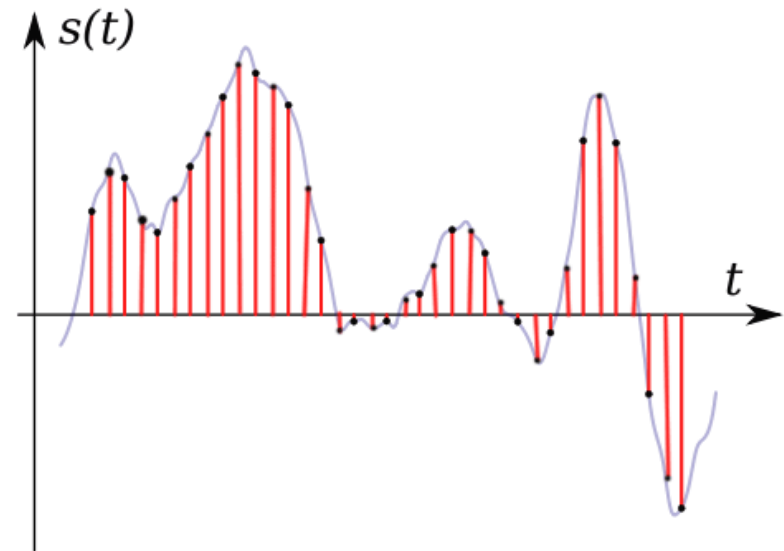
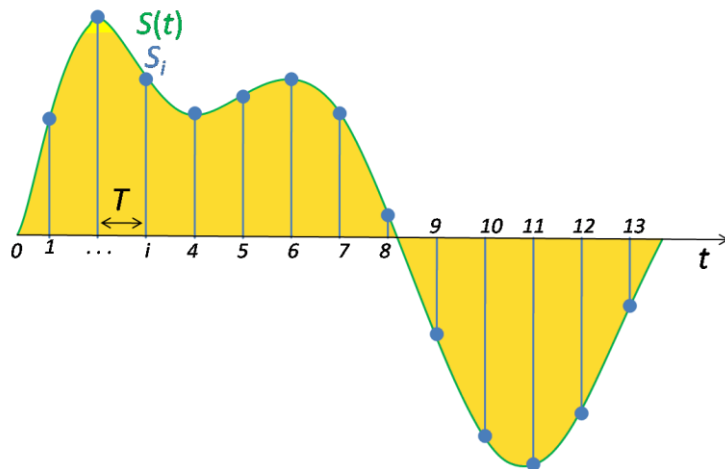
Sampling

Example

Examples of ideal sampling.

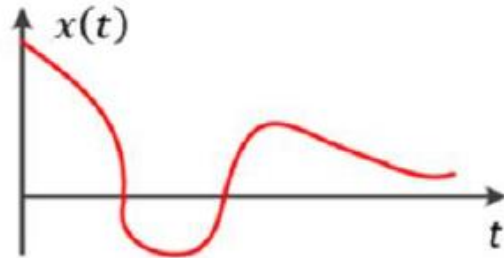
Take individual points (of infinitesimal duration) at constant-spaced intervals of time (called sampling period). Sampling frequency is its reciprocal.

In real life, we will have some non-idealities (jitter and aperture error)

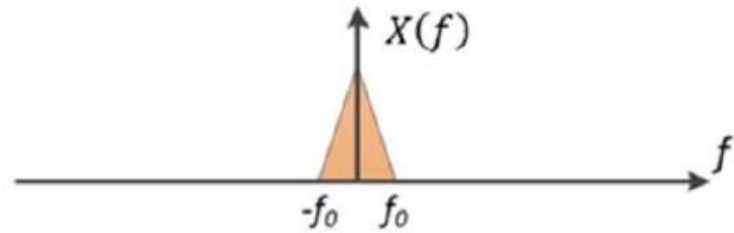


Sampling Rate

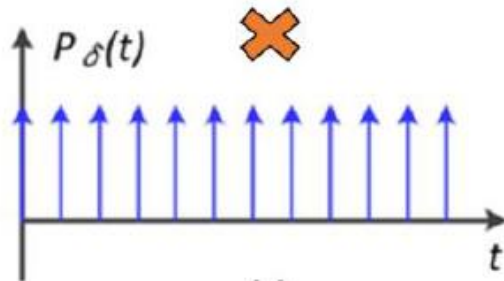
Spectrum of a sampled signal



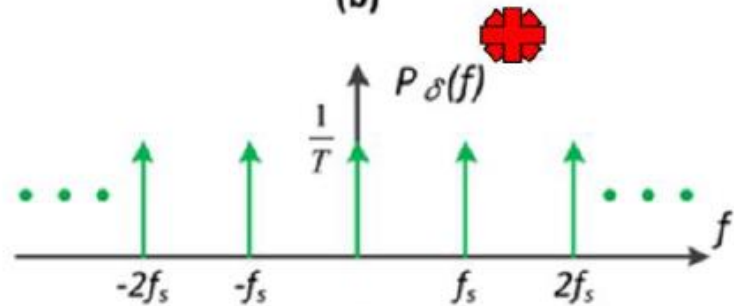
(a)



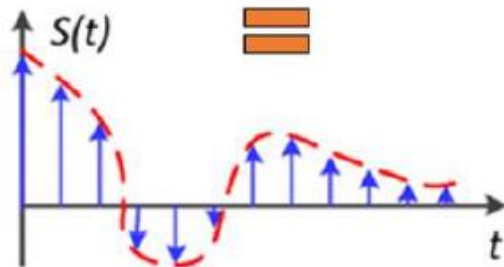
(b)



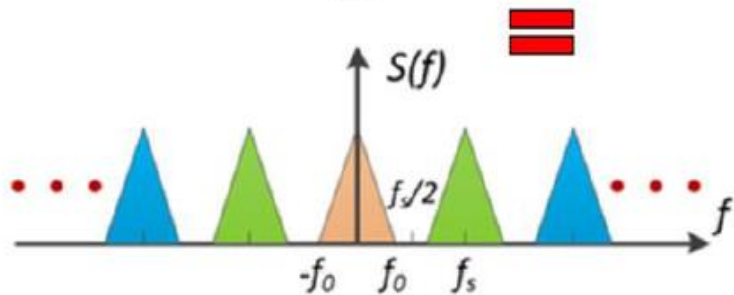
(c)



(d)



(e)



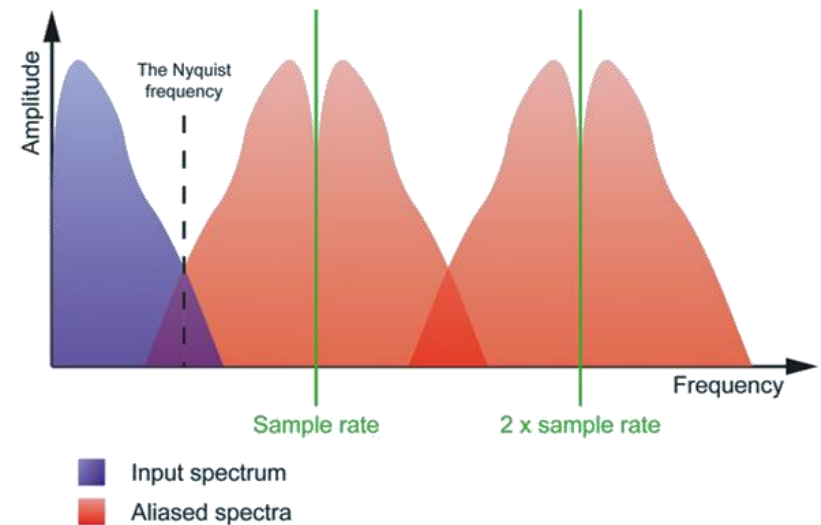
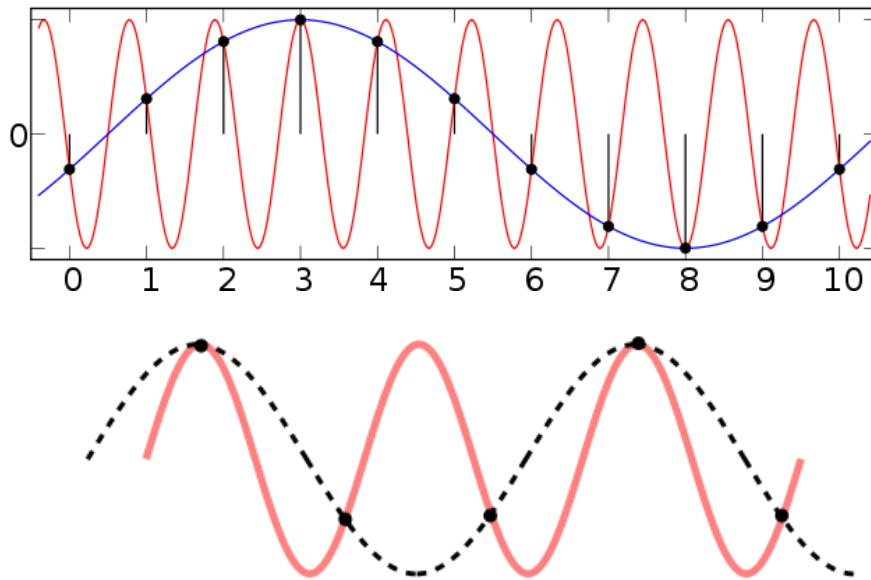
(f)

Sampling Rate

Which (signal) is which (alias)?

Aliasing is a phenomenon according to which copies of the frequency spectrum are created upon sampling (for more see Shannon-Nyquist theorem).

It causes different signals to become indistinguishable (or aliases of one another) when sampled. It also refers to the artifact that results when the signal reconstructed from samples is different from the original continuous signal. An example that is known to everyone is the wagon-wheel effect



Sampling Rate

Anti-Aliasing Filter

Aliasing must be prevented with an anti-aliasing filter

Essentially a low-pass filter used before a sampler to restrict the bandwidth of a signal to approximately or completely satisfy the Nyquist theorem over the band of interest.

Unambiguous reconstruction of the signal from its samples is possible when the power of frequencies above the Nyquist frequency is zero.

A real anti-aliasing filter trades off between bandwidth and aliasing.

A realizable anti-aliasing filter will typically either permit some aliasing to occur or else attenuate some in-band frequencies close to the Nyquist limit. In other words, aliasing cannot be completely eliminated when sampling signals that have a finite duration in time.

This is why in real systems sampling is done at 5 to 10 times the bandwidth

Quantization

Approximately

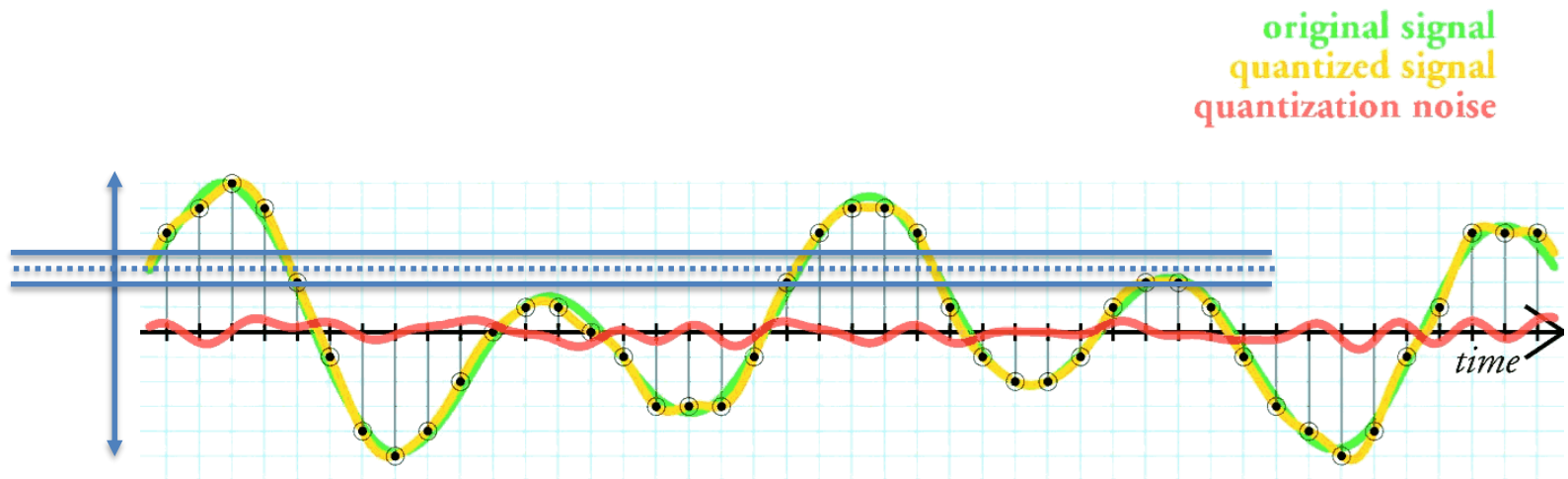
Quantization is needed to reduce precision from infinity to a finite value.

Quantization replaces each real number with an approximation from a finite set of discrete values. An n -bit quantizer will have a set of 2^n discrete values.

Typical: 8-bit (256 levels), 16-bit (65536 levels) and 24-bit (16.8 million levels).

Quantizing a sequence of numbers produces a sequence of quantization errors which is sometimes modeled as an additive random signal called quantization noise because of its stochastic behavior.

The more levels a quantizer uses, the lower is its quantization noise power.



Quantization

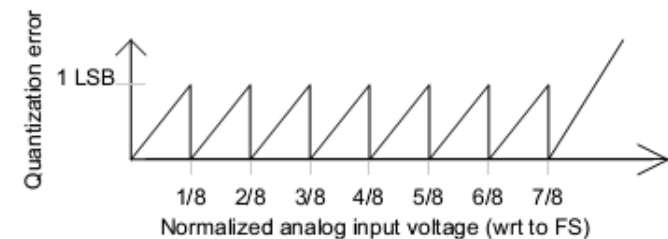
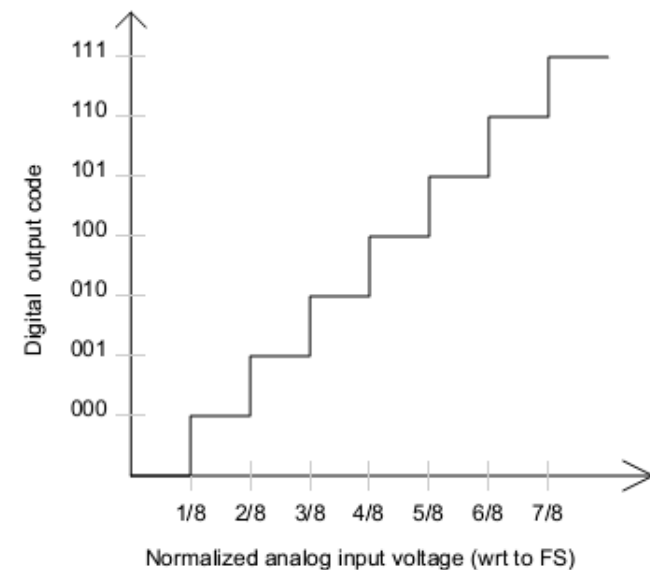
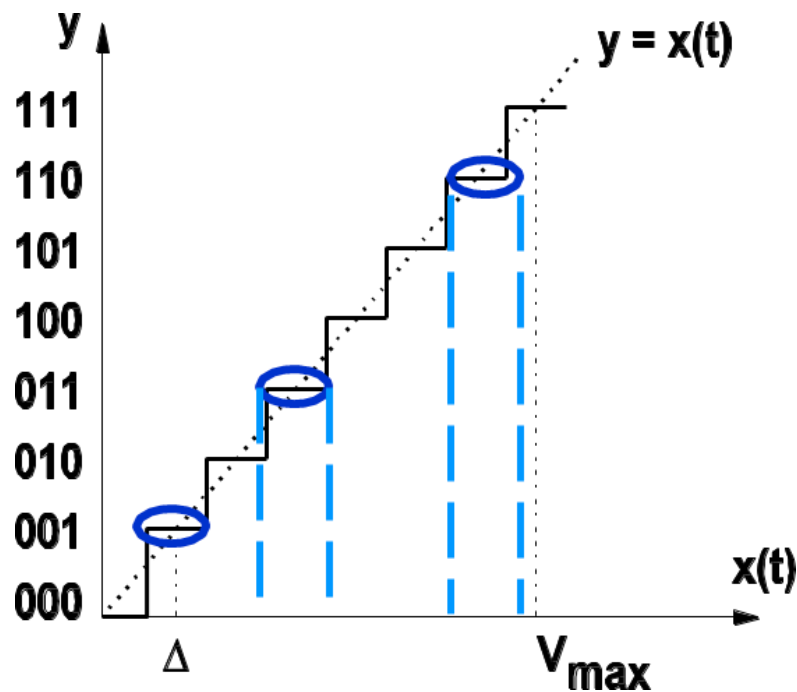
Approximately

Quantization is needed to reduce precision from infinity to a finite value.

Example: For a 3 bit A/D converter, $N=2^3=8$.

Example of an Analog quantization step size using 3 bits and a 10V wide signal:

$$Q=(V_{\max} - V_{\min})/N = (10V - 0V)/8 = 1.25V$$



Encoding

Mapping to bits

Encoding assigns the digital value (binary number) to each state finally realizing our A/D conversion.

Example of quantization range and encoding (see previous slides)

Output States	Discrete Voltage Ranges (V)
0	0.00-1.25
1	1.25-2.50
2	2.50-3.75
3	3.75-5.00
4	5.00-6.25
5	6.25-7.50
6	7.50-8.75
7	8.75-10.0

Output States	Output Binary Equivalent
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Sample and Hold

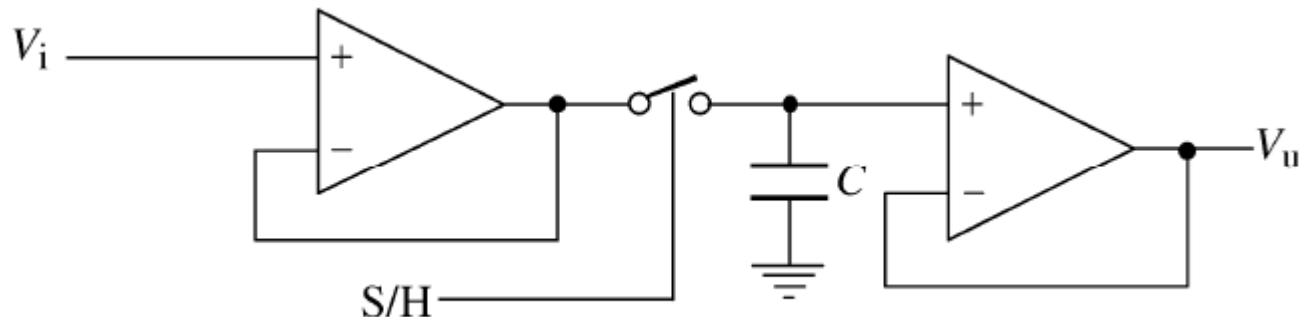
S&H Circuits

A **Sample and Hold** circuit realizes the sampling task.

The switch is controlled by a clock signal running at the sampling frequency. It selects between two modes of operation: sample and hold.

Sample: the switch is on. The input voltage is buffered and the capacitor charges up to the input voltage.

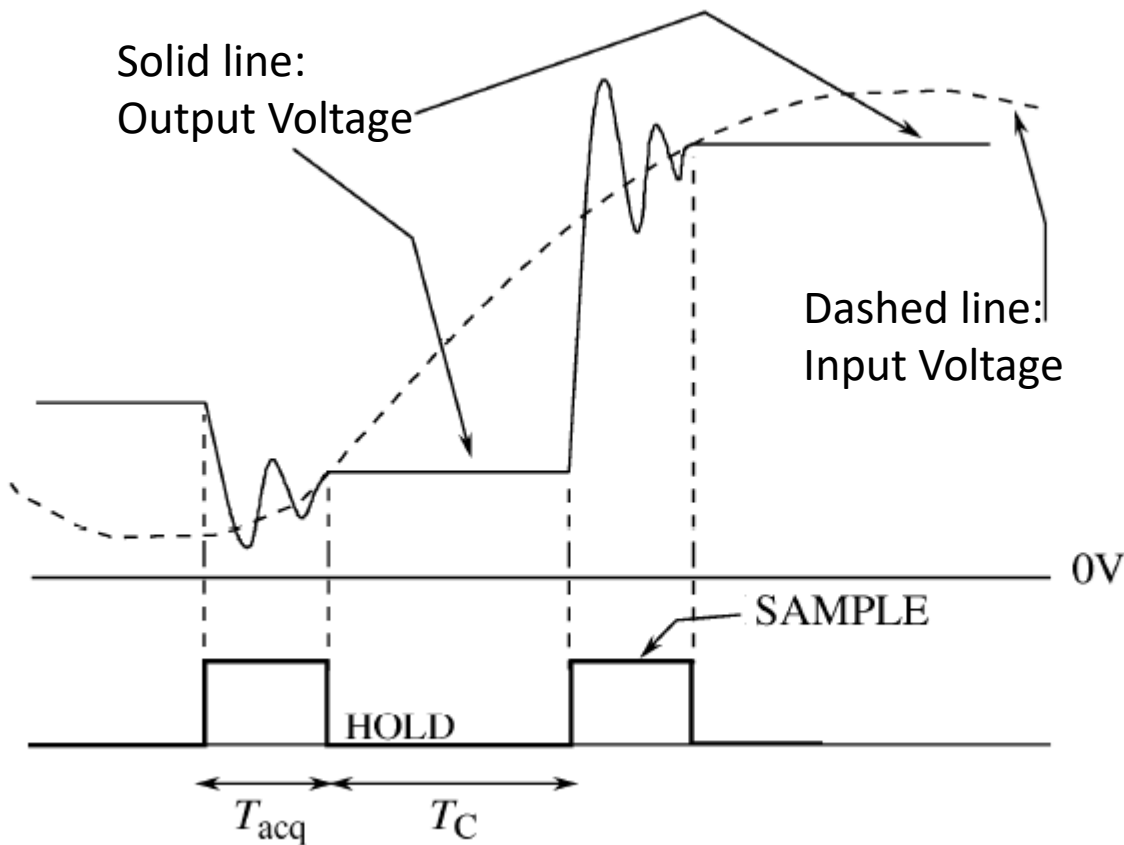
Hold: the switch is off. The capacitor holds the charge (and the voltage) as it faces the very high input impedance of the op-amp. The output voltage is fixed at the sampled value.



Sample and Hold

S&H Circuits

A **Sample and Hold** circuit realizes the sampling task.



$$T_{acq} \approx 10 \text{ ns}$$

Typologies of ADC

ADC Circuits

We will go through four kinds of ADCs – the most important and used

Flash ADC

Successive Approximation ADC

Digital-Ramp/Dual slope/Counter slope ADC (hints)

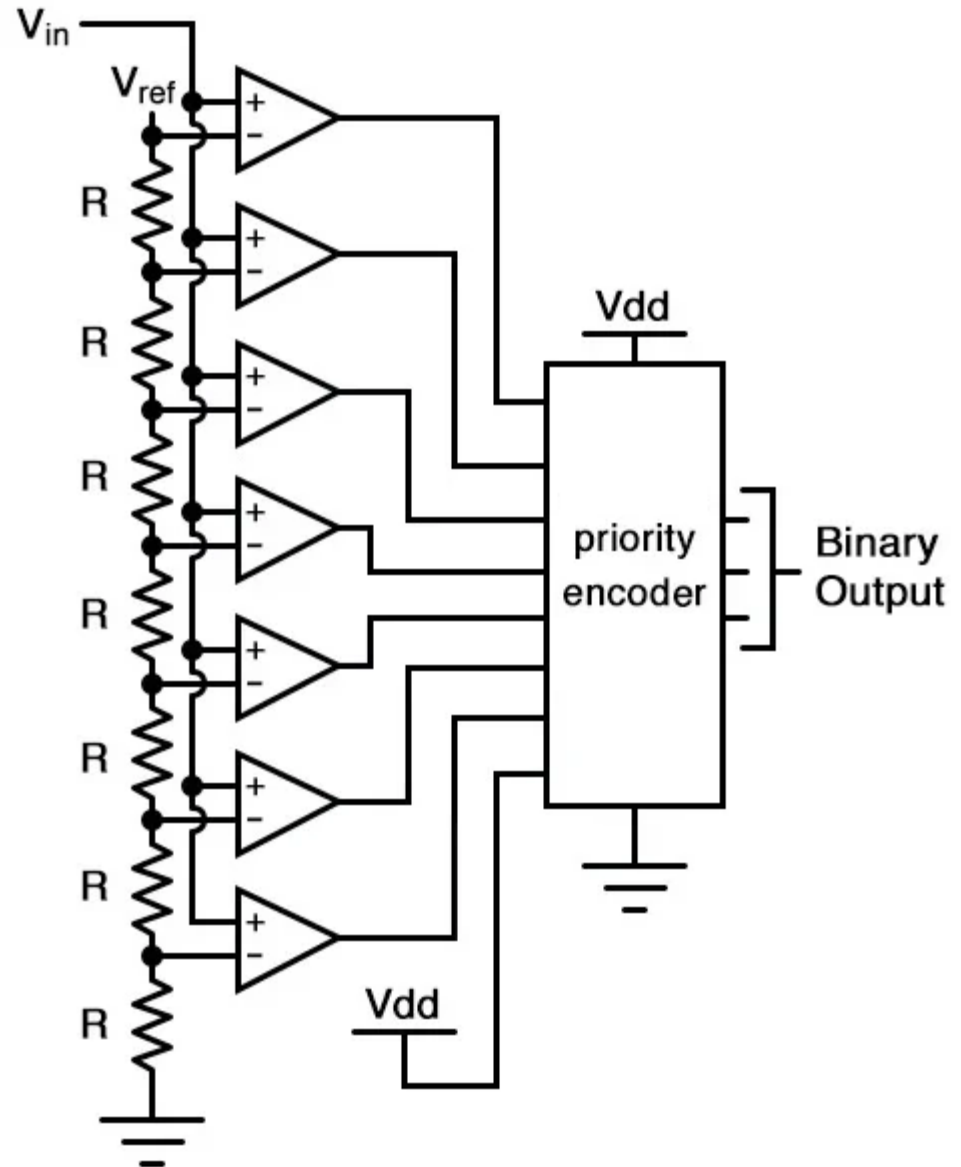
Sigma-Delta ADC (hints)

Flash ADC

ADC Circuits

Consists of a series of comparators, each one comparing the input signal to a unique reference voltage.

The comparator outputs connect to the inputs of a priority encoder circuit, which produces a binary output. As the analog input voltage exceeds the reference voltage at each comparator, the comparator outputs will sequentially saturate to a high state. The priority encoder generates a binary number based on the highest-order active input.



Flash ADC

ADC Circuits

Advantages

Simplest circuitry

Very high speed (only limited by the opamp slew rate and digital circuitry)

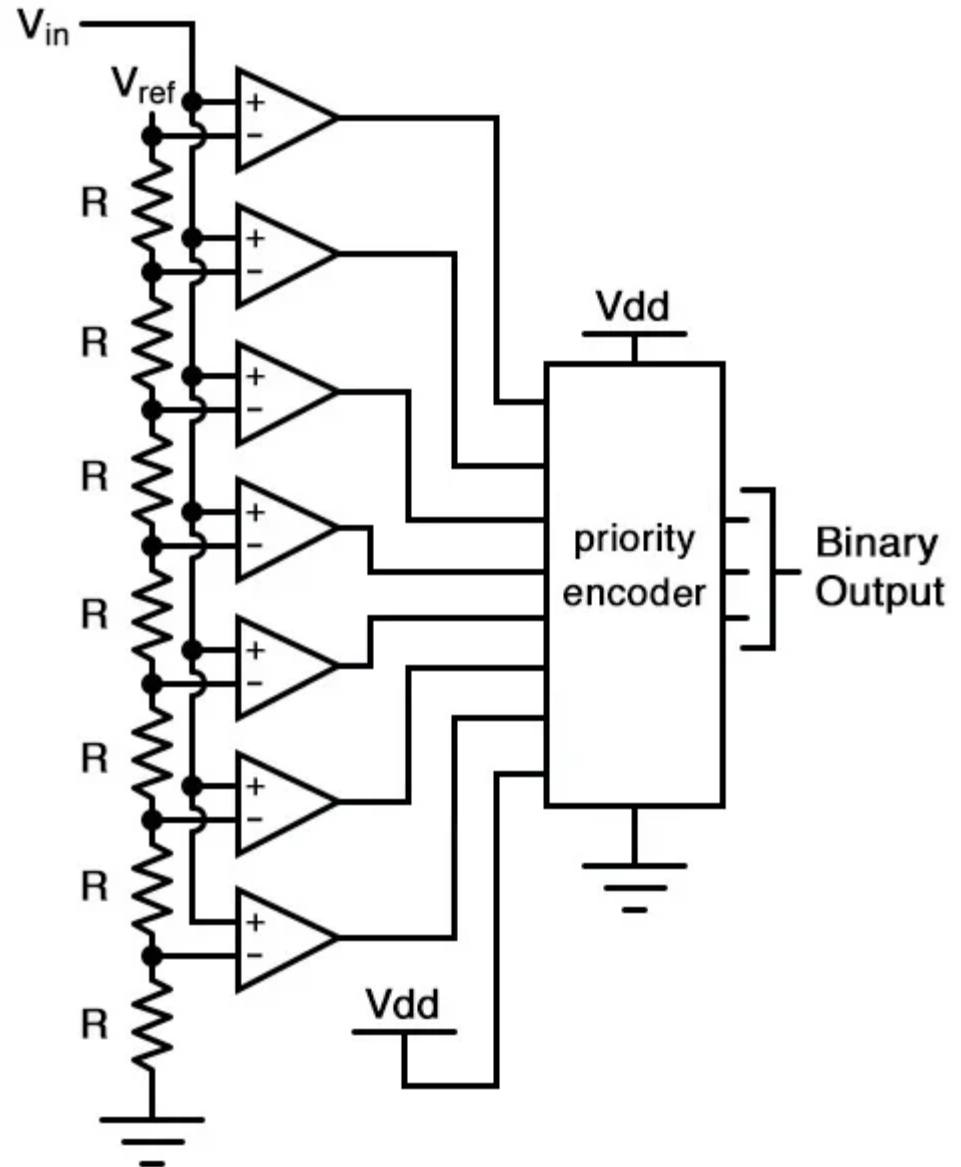
Disadvantages

Lower resolution

Expensive

For each additional output bit, the number of comparators is doubled

i.e. for 8 bits, 256 comparators needed



Successive Approximation ADC

ADC Circuits

Much faster than the dual slope ADC because it uses digital logic to converge to the digital representation that is the closest to the input voltage.

A comparator and a DAC are used in the process.

A Successive Approximation Register (SAR) is added to the circuit.

Instead of counting up in binary sequence, this register counts by trying all values of bits starting with the MSB and finishing at the LSB.

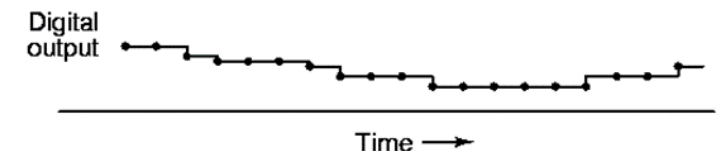
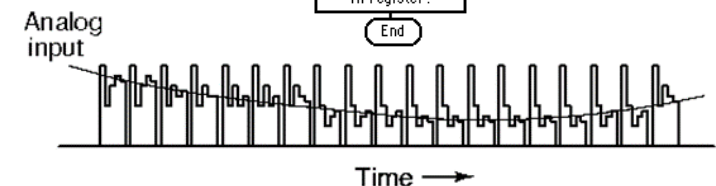
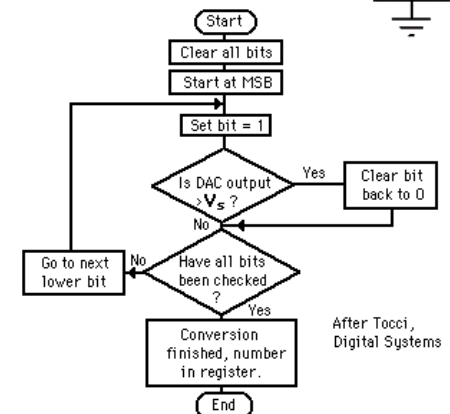
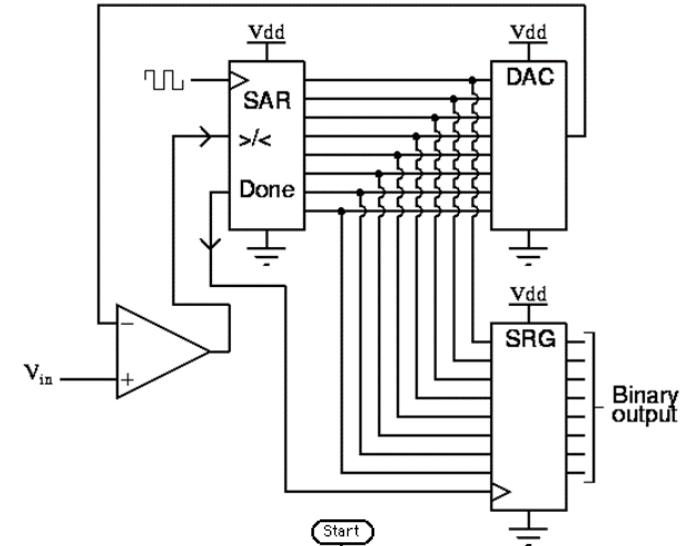
The register monitors the comparators output to see if the binary count is greater or less than the analog signal input and adjusts the bits accordingly.

Advantages

Very fast (# of bits times the clock period)

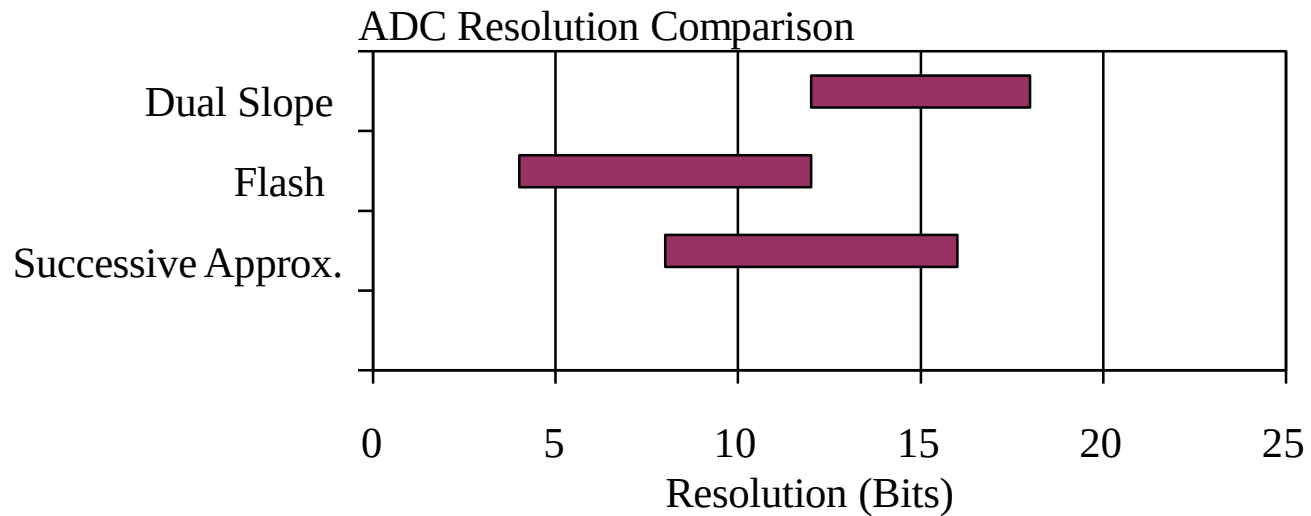
Disadvantages

SAR circuitry is complicated and requires a DAC



Comparison

ADC Circuits



Type	Speed	Cost
Dual Slope	Slow	Medium
Flash	Very Fast	High
SAR	Fast	Low

Sigma-Delta ADC

ADC Circuits

A very peculiar but much employed family of ADCs is the Sigma-Delta ($\Sigma\Delta$). They use two principles together: oversampling and noise shaping. The digital part is much more complicated while the analog part is simplified.

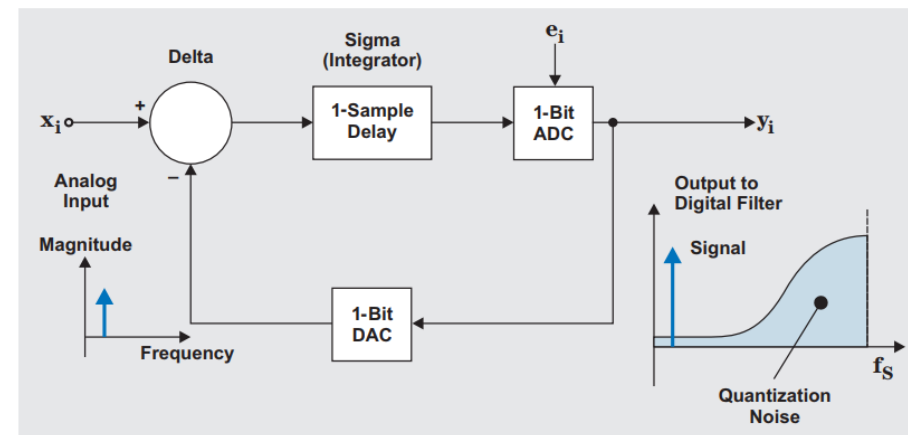
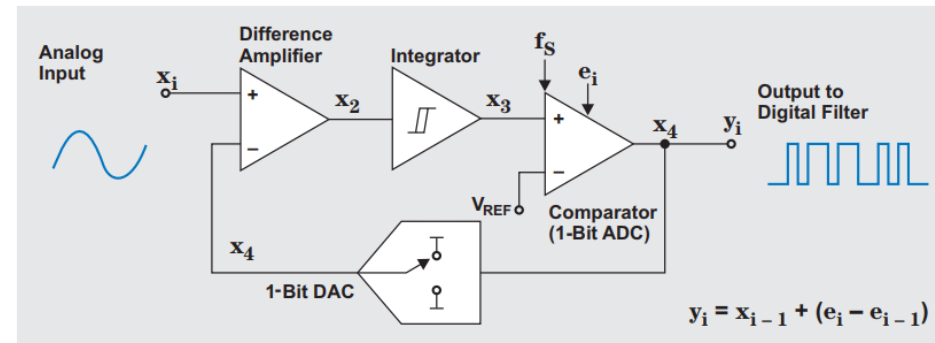
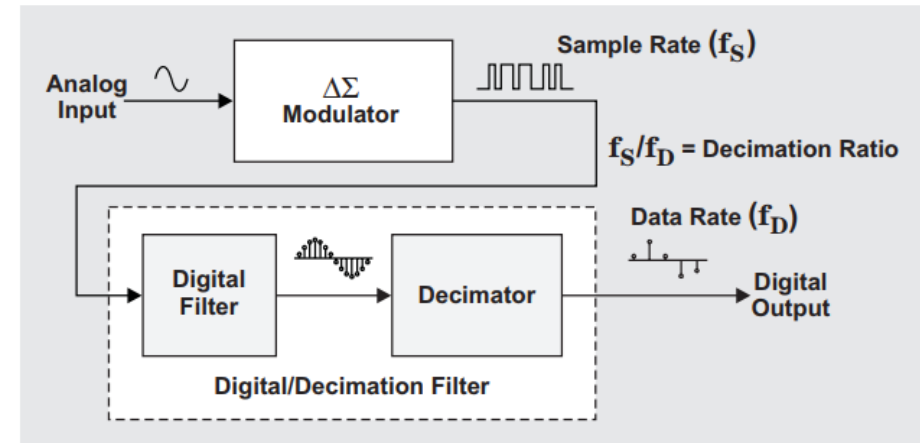
Advantages

High resolution achieved with low resolution equipment by adopting a higher sampling frequency.

Low-noise: noise power is decreased by oversampling

Disadvantages

Complicated circuitry



DAC FoMs

DAC Circuits

There are several DAC architectures; the suitability of a DAC for a particular application is determined by figures of merit including:

- Resolution
 - The number of possible output levels the DAC is designed to reproduce.
- Maximum sampling frequency
 - The maximum speed at which the DAC can produce the correct output.
- Monotonicity
 - The ability of a DAC's analog output to move only in the direction that the digital input moves (i.e., if the input increases, the output doesn't dip before asserting the correct output.) Important for low frequency signals.
- THD and Noise
 - Distortion and noise introduced by the DAC. Percentage of the total power of unwanted harmonic distortion and noise that accompany the desired signal.
- Dynamic Range
 - The difference between the largest and smallest signals the DAC can reproduce expressed in decibels.
- Power consumption, Phase Distortion, Jitter, DNL, INL

Typologies of DAC

DAC Circuits

Among many typologies of DAC circuits we will have a look at the:

- Successive Approximation DAC
- R-2R Ladder DAC (hints)
- Sigma-Delta DAC (hints)

Pulse Width Modulation is also another way

Successive Approximation DAC

DAC Circuits

A successive approximation approach to DAC. Uses mainly digital circuitry.

Advantages

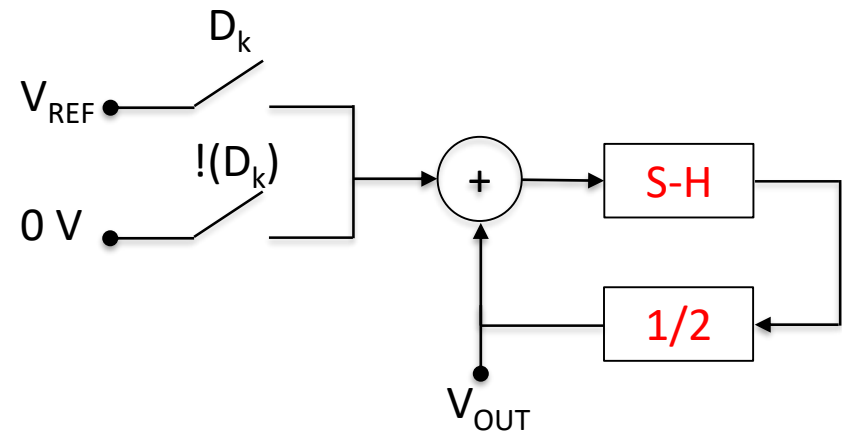
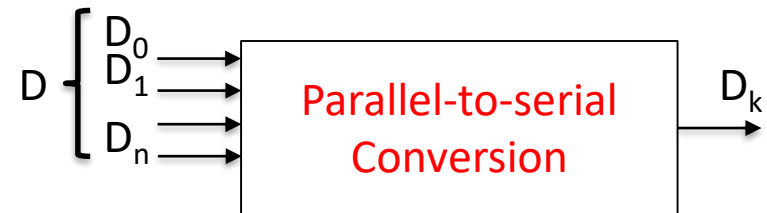
Cheap (uses mainly digital circuitry)

Precise (avoids the need for very well matched R)

Disadvantages

Less fast than R-2R

The summing node and the attenuator can be critical for precision



110

$$\begin{aligned} V_{out} &= 0V \\ V_{out} &= \frac{V_{ref}}{2} \\ V_{out} &= \frac{3V_{ref}}{4} = 7.5V \end{aligned}$$

R-2R Ladder DAC

DAC Circuits

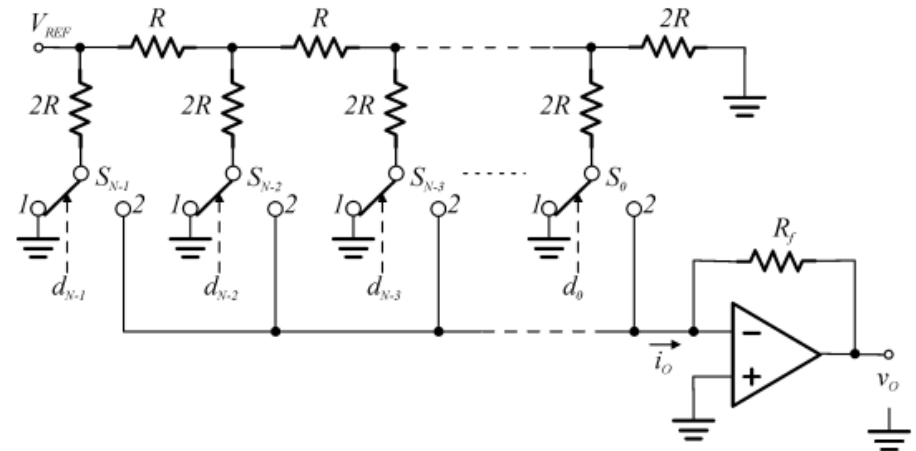
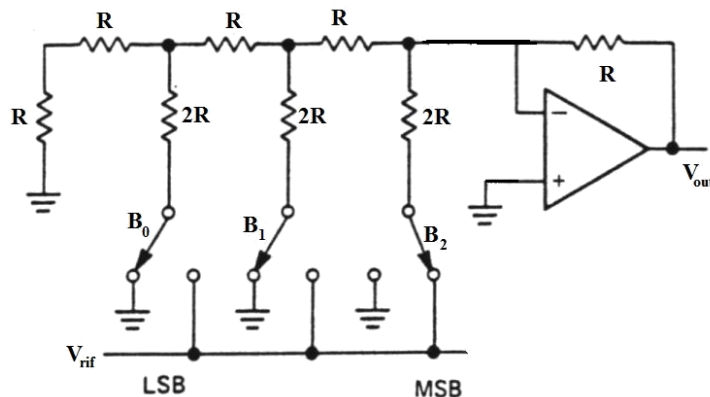
Advantages

Very fast

Only two different kinds of R required

Disadvantages

Depends critically on the R precision, on V_{REF} being REALLY constant, and on the non-idealities of both the switches and of the op-amp.



I_N is the current in the switch S_N
D is the digital input word

$$I_{N-1} = \frac{V_{REF}}{2R} = 2I_{N-2} = 4I_{N-3} = \dots = 2^{N-1}I_0$$

$$\begin{aligned} i_0 &= \frac{V_{REF}}{2R} d_{N-1} + \frac{V_{REF}}{2R} \frac{1}{2} d_{N-2} + \dots + \frac{V_{REF}}{2R} \frac{1}{2^{N-1}} d_0 = \\ &= \frac{V_{REF}}{2^N R} (d_{N-1} 2^{N-1} + d_{N-2} 2^{N-2} + \dots + d_0) = \\ &= \frac{V_{REF}}{2^N R} D \end{aligned}$$

$$v_o = -i_0 R_f = -\frac{V_{REF}}{2^N R} R_f D$$

Sigma-Delta DAC

DAC Circuits

Basically a Sigma-Delta ADC in reverse. High resolution achieved with low resolution equipment (e.g., 1-bit D/A) by adopting a higher sampling frequency.

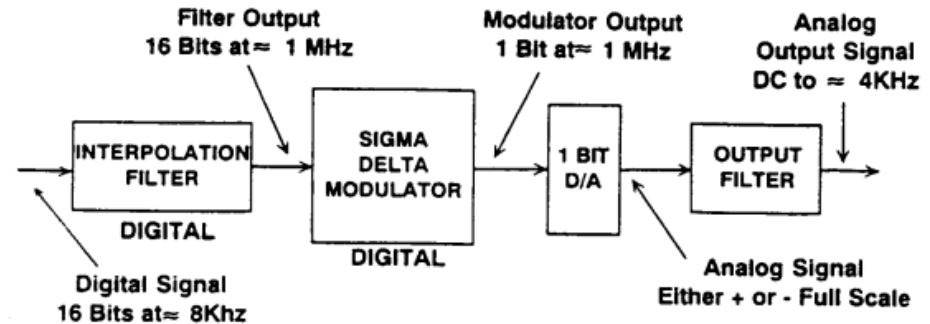
Advantages

High resolution achieved with low resolution equipment by adopting a higher sampling frequency.

Low-noise: noise power is decreased by oversampling

Disadvantages

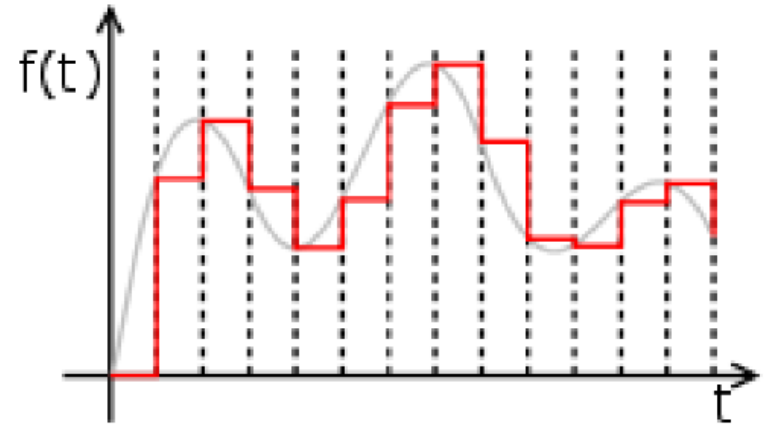
Complicated circuitry



Reconstruction Filter

ZOH

After a DAC you need something else to get your analog signal back. After all, at the output of a DAC what you get is a step-like signal that somewhat (very vaguely) resembles the original analog one.



We need to add a reconstruction filter (basically a low-pass filter at the Nyquist frequency).

The sampling theorem describes why the input of an ADC requires a low-pass analog electronic filter, called the anti-aliasing filter: the sampled input signal must be bandlimited to prevent aliasing (here meaning waves of higher frequency being recorded as a lower frequency).

For the same reason, the output of a DAC requires a low-pass analog filter, called a reconstruction filter - because the output signal must be bandlimited, to prevent imaging (meaning Fourier coefficients being reconstructed as spurious high-frequency 'mirrors').

Reconstruction Filter

ZOH

A DAC will output zero-order held that, if unfiltered, would contain spurious high-frequency content when compared to the original signal.

The reconstruction filter smooths the waveform to remove image frequencies (copies) above the Nyquist limit. In doing so, it reconstructs the continuous time signal (whether originally sampled, or modelled by digital logic) corresponding to the digital time sequence.

Ideally, both filters (reconstruction and anti-aliasing) should have:

- Constant phase delay in the pass-band
- Constant flat frequency response in the pass-band
- Zero response from the Nyquist frequency above

This could be achieved by a filter with a sinc ($\sin(x)/x$) impulse response, that however is impossible to realize. Thus, real reconstruction filters either allow some energy above the Nyquist rate, attenuate some in-band frequencies, or both.

